

1.)

$$\max z = -3x_1 + 4x_2 + x_3 + 2x_4 + 3x_5$$

$$\text{s.t.}$$

$$x_1 + x_2 + x_3 - x_4 + 2x_5 \leq 12$$

$$x_1 - 2x_2 - x_3 - x_4 - x_5 \geq -30$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

a. Standard Form

$$\text{Constraint 1: } x_1 + x_2 - x_3 - x_4 + 2x_5 \leq 12$$

$$\rightarrow \text{Slack var } s_1 \geq 0$$

$$\Rightarrow x_1 + x_2 - x_3 - x_4 + 2x_5 + s_1 = 12$$

$$\text{Constraint 2: } x_1 - 2x_2 - x_3 - x_4 - x_5 \geq -30$$

$$\rightarrow \text{Surplus var } s_2 \geq 0$$

$$\Rightarrow x_1 - 2x_2 - x_3 - x_4 - x_5 - s_2 = -30$$

$$\text{RHS non-neg} \Rightarrow -x_1 + 2x_2 + x_3 + x_4 + x_5 + s_2 = 30$$

Std:

$$\text{form } \max z = -3x_1 + 4x_2 + x_3 + 2x_4 + 3x_5$$

$$\text{s.t.}$$

$$x_1 + x_2 - x_3 - x_4 + 2x_5 + s_1 = 12$$

$$-x_1 + 2x_2 + x_3 + x_4 + x_5 + s_2 = 30$$

$$x_1, x_2, x_3, x_4, x_5, s_1, s_2 \geq 0$$

2)

a. Sketch for each $\leq$ constraint

$$\rightarrow \max z = 2x_1 + 2x_2$$

s.t.

$$x_2 + s_1 = 10$$

$$x_1 - 3x_2 + s_2 = 2$$

$$x_1 + x_2 + s_3 = 16$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

3)

a. Sketch to lines $\leq b =$

$$\rightarrow \max z = 4x_1 + x_2$$

s.t.

$$2x_1 - 3x_2 + s_1 = 12$$

$$-4x_1 + x_2 + s_2 = 8$$

$$x_1, x_2, s_1, s_2 \geq 0$$